CSGE602055 Operating Systems
CSF2600505 Sistem Operasi
Week 09: Storage, Firmware, Bootloader, & Systemd

C. BinKadal

Sendirian Berhad

https://os.vlsm.org/Slides/os09.pdf
Always check for the latest revision!

REV414: Fri 25 Aug 2023 13:00

OS232[3]): Operating Systems Schedule 2023 - 2

Week	Topic[1])	OSC10[2])
Week 00	Overview (1), Assignment of Week 00	Ch. 1, 2
Week 01	Overview (2), Virtualization & Scripting	Ch. 1, 2, 18.
Week 02	Security, Protection, Privacy, & C-language.	Ch. 16, 17.
Week 03	File System & FUSE	Ch. 13, 14, 15.
Week 04	Addressing, Shared Lib, & Pointer	Ch. 9.
Week 05	Virtual Memory	Ch. 10.
Week 06	Concurrency: Processes & Threads	Ch. 3, 4.
Week 07	Synchronization & Deadlock	Ch. 6, 7, 8.
Week 08	Scheduling + W06/W07	Ch. 5.
Week 09	Storage, Firmware, Bootloader, & Systemd	Ch. 11.
Week 10	I/O & Programming	Ch. 12.

[1]) For schedule, see https://os.vlsm.org/#idx02

[2]) Silberschatz et. al.: **Operating System Concepts**, 10th Edition, 2018.

[3]) This information will be on **EVERY** page two (2) of this course material.

STARTING POINT — https://os.vlsm.org/

- **Text Book** — Any recent/decent OS book. Eg. (**OSC10**) Silberschatz et. al.: **Operating System Concepts**, 10th Edition, 2018. (See https://www.os-book.com/OS10/).
- **Resources** (https://os.vlsm.org/#idx03)
  - **SCELE OS232** — https://scele.cs.ui.ac.id/course/view.php?id=3614. The enrollment key is **XXX**.
  - **Download Slides and Demos from GitHub.com** — (https://github.com/os2xx/os/) os00.pdf (W00), os01.pdf (W01), os02.pdf (W02), os03.pdf (W03), os04.pdf (W04), os05.pdf (W05), os06.pdf (W06), os07.pdf (W07), os08.pdf (W08), os09.pdf (W09), os10.pdf (W10).
  - **Problems** 195.pdf (W00), 196.pdf (W01), 197.pdf (W02), 198.pdf (W03), 199.pdf (W04), 200.pdf (W05), 201.pdf (W06), 202.pdf (W07), 203.pdf (W08), 204.pdf (W09), 205.pdf (W10).
  - **LFS** — http://www.linuxfromscratch.org/lfs/view/stable/
  - **OSP4DISS** — https://osp4diss.vlsm.org/
  - **This is How Me Do It!** — https://doit.vlsm.org/
    - PS: "Me" rhymes better than "I", duh!

Agenda

1. Start
2. OS232 Schedule
3. Agenda
4. Week 09
5. OSC10 (Silberschatz) Chapter 11
6. Storage, Firmware, Bootloader, & Systemd
7. Storage Management
8. RAID
9. Legacy BIOS
10. UEFI
11. Operating System (Boot) Loader
12. GRUB Map
13. init (SYSV legacy)
14. UpStart - Ubuntu
15. The All New "systemd"
16. systemctl

Week 09 Storage, Firmware, Bootloader, & Systemd: Topics[1]

- Storage
- Storage Arrays
- BIOS
- Loader
- Systemd

[1] Source: ACM IEEE CS Curricula

Week 09 Storage, Firmware, Bootloader, & Systemd: Learning Outcomes[1]

- Storage [Usage]
- Storage Arrays [Usage]
- BIOS [Usage]
- Loader [Usage]
- Systemd [Usage]

[1] Source: ACM IEEE CS Curricula

OSC10 (Silberschatz) Chapter 11

- OSC10 Chapter 11: Mass-Storage Systems
  - Overview of Mass Storage Structure
  - HDD Scheduling
  - NVM Scheduling
  - Error Detection and Correction
  - Storage Device Management
  - Swap-Space Management
  - Storage Attachment
  - RAID Structure

Storage, Firmware, Bootloader, & Systemd

- Reference: (OSC10-ch11)
- Storage Capacity (2019)[1]
  - Legacy 3.5" Floppy Disk (1.4MB) – obsolete?
    - SuperDisk (up to 240 MB) — never took off.
  - 4.7" Compact Disc (700MB) – obsolete?
    - 4.7" Digital Versatile Disc (up to 9GB) – ?
    - 4.7" Blu Ray (up to 128 GB) ⇒ DVD++.
  - Tape Cartridge (up to 15TB)
    - Robotic System (up to 250 PB per unit)
    - NASA, Google, Microsoft are still using this!
    - Cheap but slow.
  - Hard Disk Drives (up to 16 TB).
    - From Perpendicular Magnetic Recording to Shingled Magnetic Recording technology (+25% – writing problems).
    - Mechanical Disk Arm Scheduling (Until When?).
  - Solid-State Disks (up to 16 TB).
    - SSD Price > HDD Price.
    - Write Speed >> Read Speed.
    - (What is a) Flash Disk?

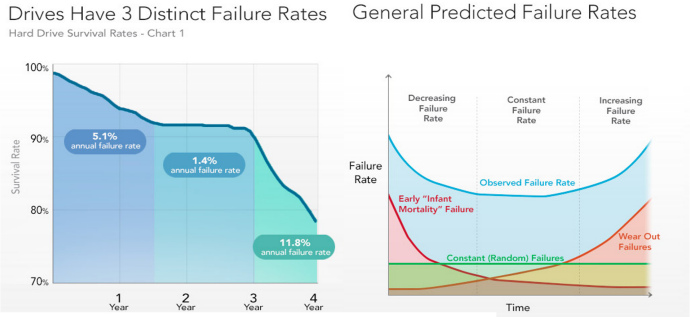
[1]

Storage Capacity

Figure: Source: https://linktr.ee/hackingarticles

Storage Failure Rates

- MTTDL: Mean Time To Data Loss
- MTTF: Mean Time To Failure
- BackBlaze (Cloud Backup Services)



https://www.extremetech.com/computing/170748-how-long-do-hard-drives-actually-live-for

BACKBLAZE

Figure: BackBlaze — Failure Rates of 25000 DISKS

Storage Management

- Attached-Storage.
  - Host-Attached Storage: via I/O.
  - Network-Attached Storage (NAS): via distributed FileSystem.
  - Storage Area Network (SAN): dedicated Network.
- Formating
  - Low Level (Physical)
  - High Level (FileSystem)
- Boot Block
- Disk Partition
  - "MBR"-scheme
    - upto 4 primary partition
    - upto 2 TB disk
  - "GPT"-scheme
    - "unlimited" partition
    - "unlimited" disk
    - redundancy
- Swap Space Management: On Partition or FileSystem?

RAID: Redundant Array of In* Disks

- RAID 0, 1, 5, 6, 10, 100
- Note (http://www.commodore.ca/windows/raid5/raid5.htm):
  - RAID was created to enhance data performance, reliability and availability.
  - Striping, parity checking and mirroring are three primary functions of RAID systems.
  - RAID performs its functions transparent to the operating system.
  - Systems are typically defined by ranks consisting of five disks each connected to one or two Disk Array Controllers.
  - Different RAID levels provide varying degrees of speed and data protection.
- Problems with RAID
- Stable-Storage Implementation

BIOS, Boot, & Systemd

- Firmware
  - BIOS: Basic Input Output System.
  - UEFI: Unified Extensible Firmware Interface.
  - ACPI: Advanced Configuration and Power Interface.
- Operating System (Boot) Loader
  - BOOTMGT: Windows Bootmanager / Bootloader.
  - LILO: Linux Loader.
  - GRUB: GRand Unified Bootloader.
- Operating System Initialization
  - Init (legacy)
  - UpStart
  - Systemd

Legacy BIOS

- Check Settings.
- Initialize CPU & RAM.
- POST: Power-On Self-Test.
- Initialize ports, LANS, etc.
- Load a Boot Loader.
- Handover to the Boot Loader.
- Provides "Native" (obsolete) Drivers only (not loadable).
- Provides "INT" services .
- Limitation.
  - Technology of 1970s.
  - 16 bits software.
  - 20 bits address space (1 MB).
  - 31 bits disk space (2 TB).

BIOS

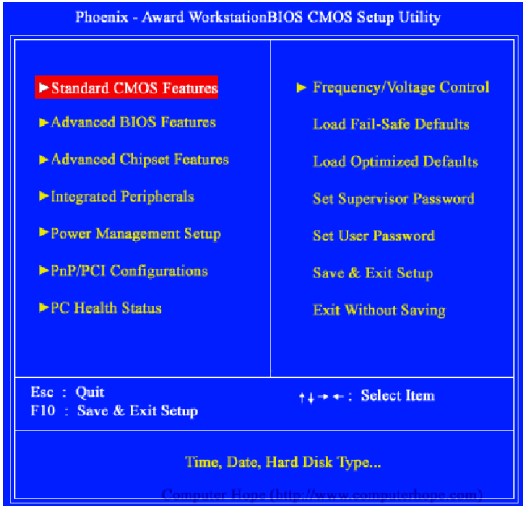


Figure: BIOS

UEFI

- A Firmware Specification, not an Implementation!
- No (INT) service after boot.
- HII: Human Interface Infrastructure.
- Protected Mode.
- Flexible.
  - Technology of 2000s.
  - writen in C.
  - (third party) loadable drivers and tools.
  - Emulate Legacy BIOS transition (MBR block, INT service).
  - UEFI Shell: environment shell for diagnostic (no need for DOS).
- Problems
  - Who controls the Hardware?
  - Is "Secure Boot" a good thing?
  - How about a **NASTY/LOCKING/TROJAN** UEFI implementation?
  - Different **DRIVERS**.

UEFI

Figure: UEFI

Platform Initialization (PI) Boot Phases

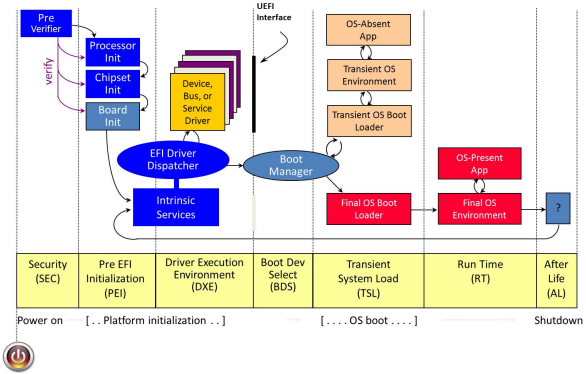


Figure: UEFI Boot Process[1].

[1] Source Jarslstrom - 2014 - www.tianocore.org

Operating System (Boot) Loader

- General
  - How/Where to start the operating system?
  - What to do?
  - How many ways to boot?
  - How many types of OS?
- Disk Partition
  - MBR: Master Boot Record (1983).
  - GPT: GUID (Globally Unique Identifiers) Partition Table (2010s).
- GRUB: GRand Unified Boot system
  - Stage 1: a small boot.img inside the MBR.
  - Stage 1.5 (core.img): FileSystem drivers after MBR.
  - Stage 2: Kernel Selection: Windows, Linux, BSD, etc.
- GRUB2
  - More flexible than GRUB legacy.
  - More automated than GRUB legacy.
  - Accept MBR and GPT.
  - Stage 1.5 (core.img): generated from diskboot.img.
  - No 1024 cylinder restriction.

GRUB Map

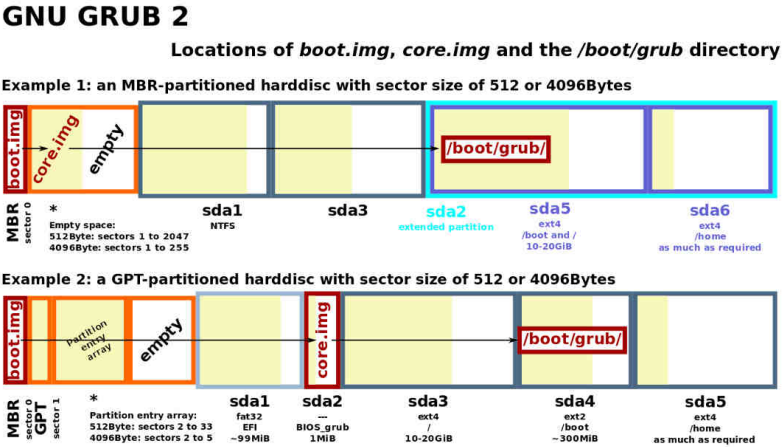


Figure: GRUB[1].

[1] Source Shmuel Csaba Otto Traian 2013

init (SYSV legacy)

- File: `/etc/inittab`.
- Folders: `/etc/rcX.d` — X = runlevel.
  - Seven (7) different runlevels:
    - 0 (shutdown).
    - 1 (single-user/admin).
    - 2 (multi-user non net).
    - 3 (standard).
    - 4 (N/A).
    - 5 (3+GUI).
    - 6 (reboot).
  - SXX-YYY: Start
  - KXX-YYY: Kill.
- One script at a time in order.
- dependency is set manually.

UpStart - Ubuntu

- Developer: Ubuntu.
- Folder: `/etc/init/`.
- Control: `initctl`.
  - `initctl list` – listing all processes managed by upstart.
- better support for hotplug devices.
- cleaner service management.
- faster service management.
- asynchronous.

The All New "systemd"

- Replaces (SYSV) init and UpStart.
  - better concurency handling: Faster!
  - better dependencies handling: No more "S(tarts)" and "K(ills)".
  - better crash handling: automatic restart option.
  - better security: group protection from anyone including superusers.
  - simpler config files: reliable and clean scripts.
  - hotplug: dynamic start/stop.
  - supports legacy systems (init).
  - overhead reducing.
  - unified management way for all distros.
  - bloated: doing more with more resources.
  - linux specific: NOT portable.

systemctl 01

```
for II in      \
   'systemctl list-unit-files | head -8; echo "(...)";
       systemctl list-unit-files| tail -8' \
   'systemd-analyze blame | wc -l; echo "===";
       systemd-analyze blame | head -15' \
   'systemctl --full  | wc -l; echo "===";
       systemctl --full  | head -10' \
   'systemctl list-units  | wc -l; echo "===";
       systemctl list-units  | head -10' \
   'systemctl list-units |grep .service|wc -l;echo "===";
       systemctl list-units|grep .service|head -10' \
   'systemctl list-units  | grep ssh.service' \
   'systemctl status ssh.service' \
   'systemctl is-enabled ssh' \
   'journalctl' \
   'journalctl -b' \
do
```

systemctl 02

```
$ bash a1-some-systemd-command-lines

This is a SYSTEMD - systemctl demo.
Just run: "bash a1-some-systemd-command-lines"

*** Hit Enter Key ***
```

Figure: bash a1-some-systemd-command-lines

systemctl 03

```
rms46@pamulang1: ~
File  Edit  View  Search  Terminal  Tabs  Help
root@pamulang1: ~ | rms46@pamulang... | rms46@pamulang... | @jembatan: ~ | rms46@pamulang... | rms46@pamulang... | rms46@pamulang... | rms46@pamulang... | rms46@pamulang... | rms46@pamulang...
================================================================================
RUNNING: systemctl list-unit-files | head -8; echo "(...)";systemctl list-unit-files| tail -8
================================================================================
UNIT FILE                                STATE
proc-sys-fs-binfmt_misc.automount        static
dev-hugepages.mount                      static
dev-mqueue.mount                         static
proc-sys-fs-binfmt_misc.mount            static
sys-fs-fuse-connections.mount            static
sys-kernel-config.mount                  static
sys-kernel-debug.mount                   static
(...)
time-sync.target                         static
timers.target                            static
umount.target                            static
mdadm-last-resort@.timer                 static
systemd-readahead-done.timer             static
systemd-tmpfiles-clean.timer             static

223 unit files listed.

*** Hit Enter Key ***
```

Figure: systemctl list-unit-files

systemctl 04

```
rms40@pamulang1: ~
File  Edit  View  Search  Terminal  Tabs  Help
root@pamulang1: ~   rms40@pamulang...   rms40@pamulang...   @jembatan: ~   rms40@pamulang...   rms40@pamulang...   rms40@pamulang...   rms40@pamulang...   rms40@pamulang...   rms40@pamulang...
================================================================================
RUNNING: systemd-analyze blame | wc -l; echo "===";systemd-analyze blame | head -10
================================================================================
34
===
          2.374s keyboard-setup.service
           963ms systemd-logind.service
           957ms rsyslog.service
           954ms ssh.service
           954ms rc-local.service
           954ms systemd-user-sessions.service
           928ms postfix.service
           589ms networking.service
           519ms snmpd.service
           322ms systemd-tmpfiles-setup-dev.service

*** Hit Enter Key ***
```

Figure: systemd-analyze blame

systemctl 05

```
rms40@pamulang1: ~
File  Edit  View  Search  Terminal  Tabs  Help
root@pamulang1: ~ | rms40@pamulang... | rms40@pamulang... | @jembatan: ~ | rms40@pamulang... | rms40@pamulang... | rms40@pamulang... | rms40@pamulang... | rms40@pamulang... | rms40@pamulang...
==================================================================
RUNNING: systemctl --full | wc -l; echo "===";systemctl --full | head -6
==================================================================
97
===
UNIT                                                                    LOAD   ACTIVE SUB       DESCRIPTION
proc-sys-fs-binfmt_misc.automount                                       loaded active waiting   Arbitrary Executable File Formats File System Automount Point
sys-devices-pci0000:00-0000:00:05.0-host0-target0:0:0-0:0:0:0-block-sda-sda1.device   loaded active plugged   QEMU_HARDDISK 1
sys-devices-pci0000:00-0000:00:05.0-host0-target0:0:0-0:0:0:0-block-sda-sda2.device   loaded active plugged   QEMU_HARDDISK 2
sys-devices-pci0000:00-0000:00:05.0-host0-target0:0:0-0:0:0:0-block-sda.device        loaded active plugged   QEMU_HARDDISK
sys-devices-pci0000:00-0000:00:05.0-host0-target0:0:1-0:0:1:0-block-sdb-sdb1.device   loaded active plugged   QEMU_HARDDISK 1

*** Hit Enter Key ***

 ==================================================================
 RUNNING: systemctl list-units | wc -l; echo "===";systemctl list-units | head -6
 ==================================================================
 97
 ===
 UNIT                                                                   LOAD   ACTIVE SUB       DESCRIPTION
 proc-sys-fs-binfmt_misc.automount                                      loaded active waiting   Arbitrary Executable File Formats File System Automount Point
 sys-devices-pci0000:00-0000:00:05.0-host0-target0:0:0-0:0:0:0-block-sda-sda1.device   loaded active plugged   QEMU_HARDDISK 1
 sys-devices-pci0000:00-0000:00:05.0-host0-target0:0:0-0:0:0:0-block-sda-sda2.device   loaded active plugged   QEMU_HARDDISK 2
 sys-devices-pci0000:00-0000:00:05.0-host0-target0:0:0-0:0:0:0-block-sda.device        loaded active plugged   QEMU_HARDDISK
 sys-devices-pci0000:00-0000:00:05.0-host0-target0:0:1-0:0:1:0-block-sdb-sdb1.device   loaded active plugged   QEMU_HARDDISK 1

 *** Hit Enter Key ***


 ==================================================================
 UNNING: systemctl list-units |grep .service|wc -l;echo "===";systemctl list-units|grep .service|head -6
 ==================================================================
 12
 ===
 acct.service                                                                   loaded active exited    LSB: process and login accounting
 cpid.service                                                                   loaded active running   ACPI event daemon
 console-setup.service                                                          loaded active exited    Set console font and keymap
 ron.service                                                                    loaded active running   Regular background program processing daemon
 bus.service                                                                    loaded active running   D-Bus System Message Bus
 getty@tty1.service                                                             loaded active running   Getty on tty1

 *** Hit Enter Key ***
```

Figure: systemctl –full; systemctl list-units

systemctl 06

```
rms46@pamulang1: ~
File  Edit  View  Search  Terminal  Tabs  Help
root@pamulang1: ~ | rms46@pamulang... | rms46@pamulang... | @jembatan: ~ | rms46@pamulang... | rms46@pamulang... | rms46@pamulang... | rms46@pamulang... | rms46@pamulang... | rms46@pamulang...
=====================================================
RUNNING: systemctl list-units | grep ssh.service
=====================================================
ssh.service                 loaded active running   OpenBSD Secure Shell server
*** Hit Enter Key ***
=======================================
RUNNING: systemctl status ssh.service
=======================================
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled)
   Active: active (running) since Sun 2020-04-26 03:00:24 WIB; 3h 33min ago
  Process: 653 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
 Main PID: 686 (sshd)
   CGroup: /system.slice/ssh.service
           ├─ 686 /usr/sbin/sshd -D
           ├─3247 sshd: demo [priv]
           ├─3253 sshd: demo@pts/0
           ├─3254 -bash
           ├─3391 bash a1-some-systemd-command-lines
           └─3550 systemctl status ssh.service
*** Hit Enter Key ***
```

Figure: systemctl status ssh.service

systemctl 07

```
==========================================
RUNNING: systemctl is-enabled ssh
==========================================
enabled

*** Hit Enter Key ***
```

Figure: systemctl is-enabled ssh